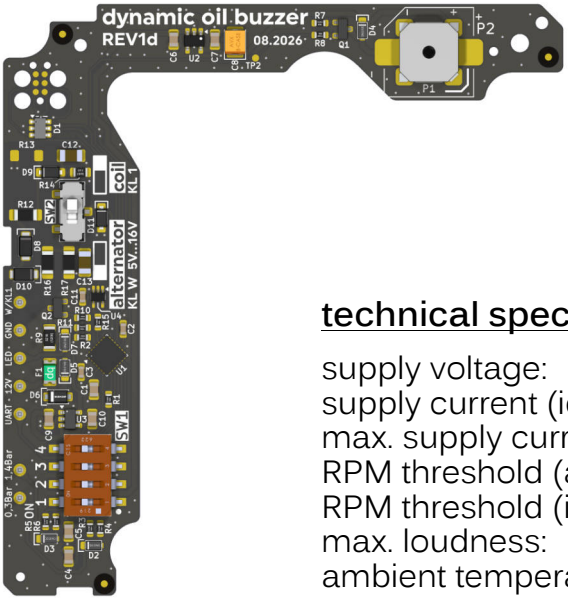
dynamic oil buzzer



github
dynamic oil buzzer

technical specifications

supply voltage:	6V ... 16 V
supply current (idle):	5 mA
max. supply current:	150 mA
RPM threshold (active):	>2050 RPM
RPM threshold (inactive):	<2000 RPM
max. loudness:	~90dBa
ambient temperature:	-40°C ... 85°C

This circuit board serves as a dynamic oil pressure monitor for diesel and benzin/petrol/gasoline engines.
When the engine speed exceeds **2050 RPM**, oil pressure 2 is checked (this function is inactive when the engine speed is below 2000 RPM). Furthermore, the circuit board beeps and flashes if the engine speed is lost despite oil pressure 2 being present (V-belt failure in diesel engines).
If oil pressure 1 is lost, the LED flashes immediately. If another fault occurs, this is signaled after 2 seconds (LED and buzzer).

If a speed above 9000 rpm is detected, the buzzer will beep and the LED will stay on! (incorrect motor setting/gear ratio).

compatibility:

T3 / vanagon
Golf 1 Cabriolet
Golf 2
Scirocco 2
Caddy 1
Jetta 2
Passat B2
Polo 2

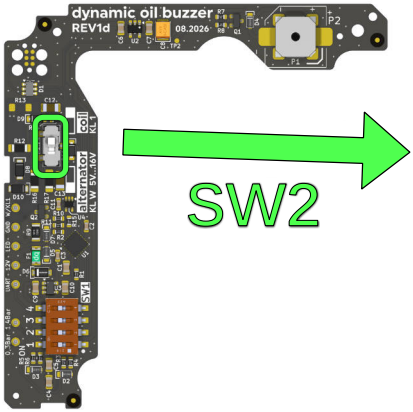
https://youtu.be/VGgJPHZ6h68

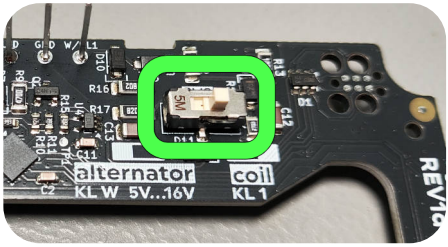
function overview and mounting instructions

select signal source

The circuit board accepts either a **5V–16V** signal (sine wave, square wave, pulse, etc.) or the **terminal 1** pulse from the ignition coil (Digifant, Digijet, carburetor).
Please note that the switch must be set correctly before installation!

Using the wrong setting can damage the electronics!





example:
switch position right „COIL“

switch position left



alternator / terminal W

- Alternator
- 5V / 12V square wave
- 5V ... 16V signal
(sine wave, square wave, pulse, etc.)

switch position right

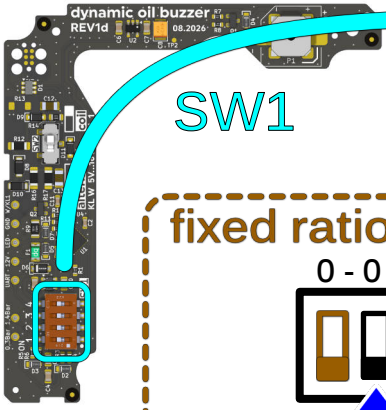


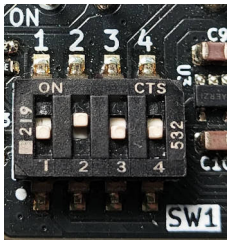
coil / terminal 1

- Ignition coil

set the engine type and pulley ratio

A 4-position switch is used to set the engine type, as well as the pulley ratio for diesel engines or the number of cylinders for gasoline engines.
In learning mode, any signal can be programmed and used based on a user-defined idle speed.



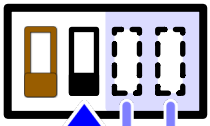


example 0-1-0-0:

- fixed ratio
- benzin
- 4 cylinder

fixed ratio

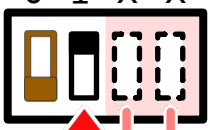
0 - 0 - X - X



diesel

00	ratio 2,01
01	ratio 2,2
10	ratio 2,39
11	ratio 2,75

0 - 1 - X - X



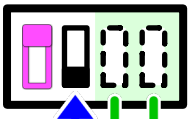
benzin

00	4 cylinder
01	5 cylinder
10	6 cylinder
11	8 cylinder

learning mode

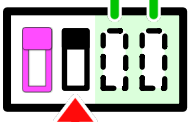
1 - 0 - X - X

diesel



1 - 1 - X - X

benzin



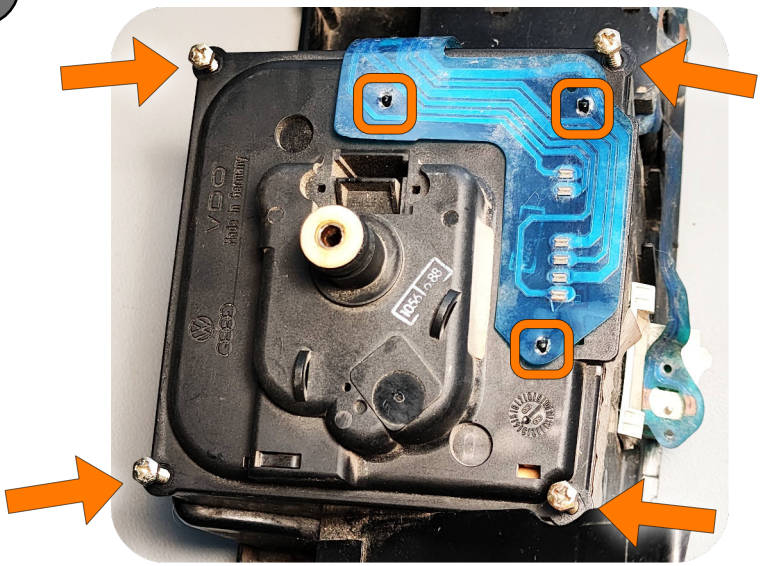
idle-RPM

00	750 rpm
01	800 rpm
10	850 rpm
11	900 rpm

Mounting instructions

Once all the DIP switches have been set correctly, the installation can begin.

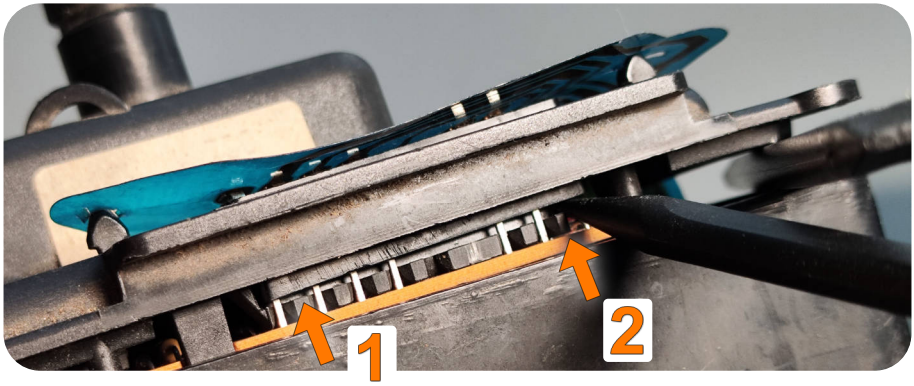
1 Removal of the conductive foil



To remove the connector as carefully as possible, you must first remove all four screws from the speedometer housing. Then carefully unclip the speedometer cover (use a plastic pry bar!).



2 Lift the circuit board connector

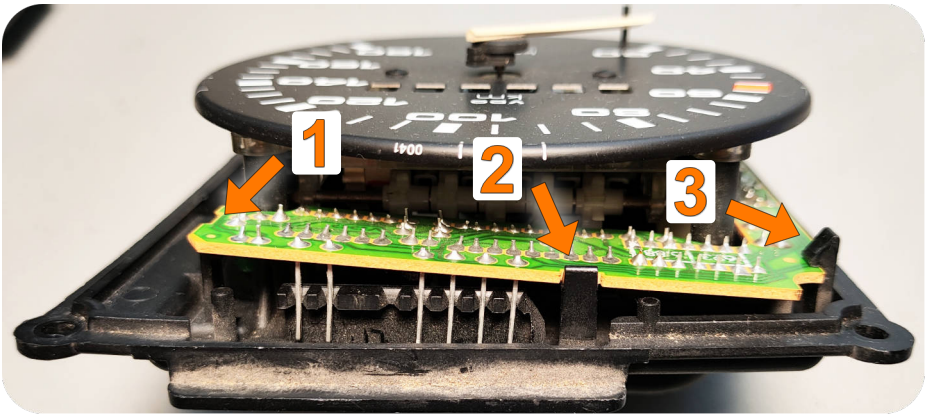


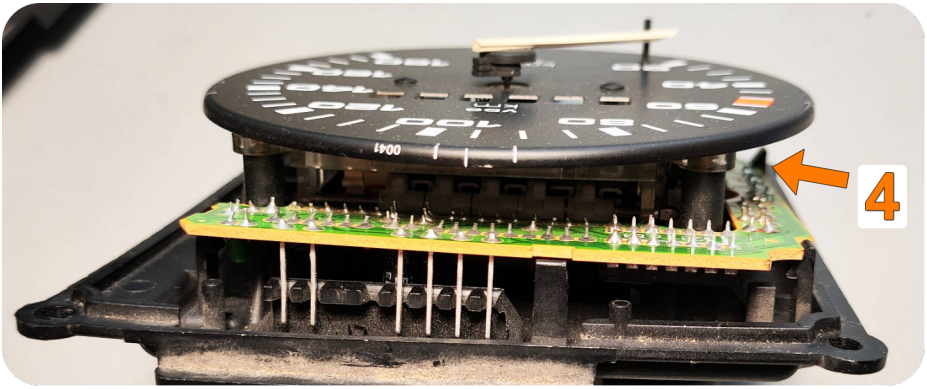
Now lift the plug carefully, alternating between each side.

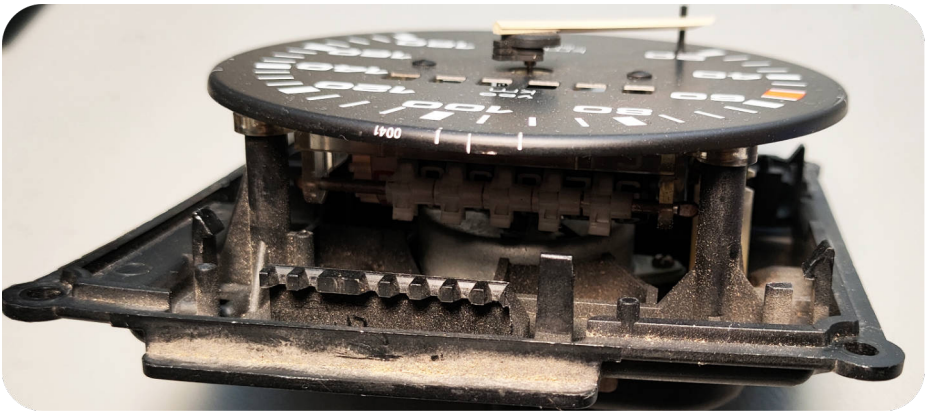




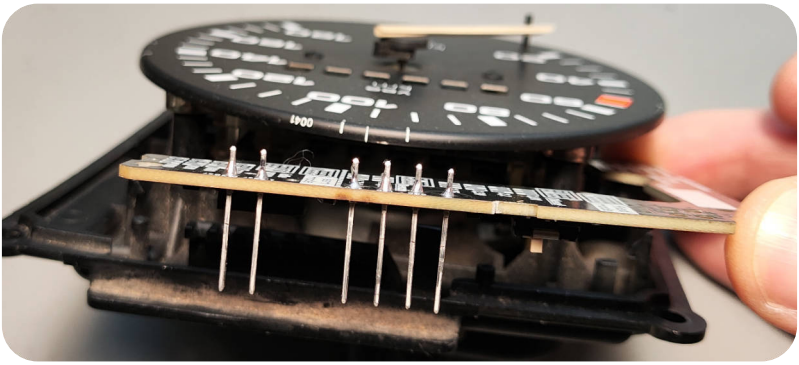
3 Unclip the circuit board

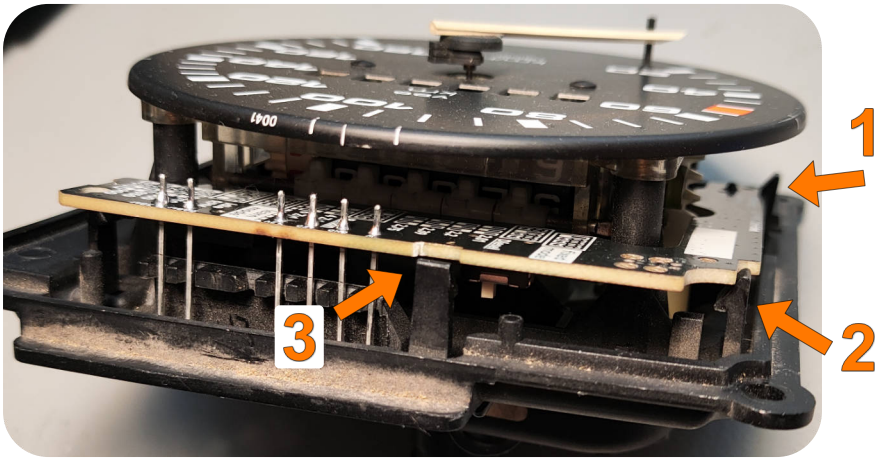


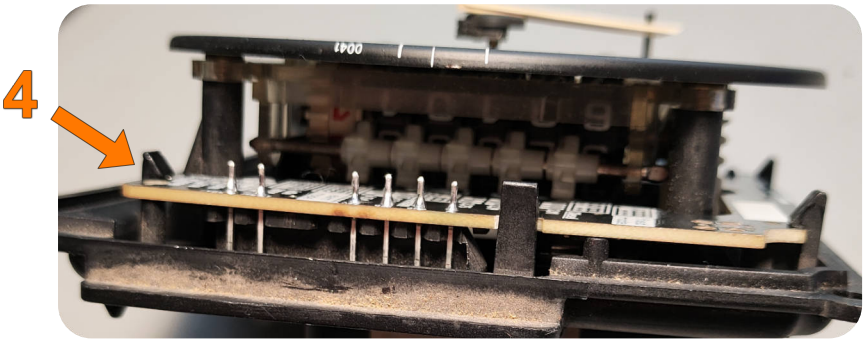




4 Clip in the new circuit board







Now screw the speedometer back into the instrument cluster, plug in the connector and fit the instrument cluster.

Commissioning



switch position: **fixed ratio**

Once installed, the car can be operated as normal.



switch position: **learning mode**

As shipped, the learning mode will be executed at first startup when the ignition is switched on.

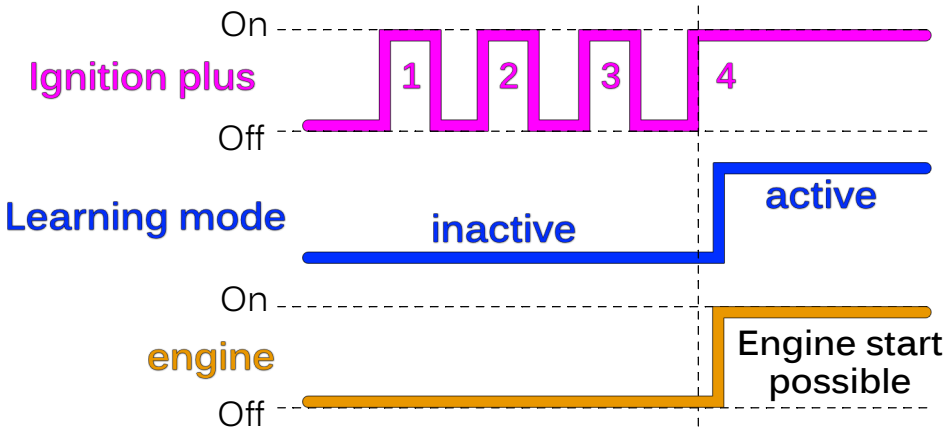
To enter learning mode, turn the ignition on **4 times** without starting the engine.

If the engine starts before the 4th attempt, the counter will reset to 0.

If the switch position has changed since the last time the unit was powered on, the learning mode will also start when the ignition is switched on.

If you wish to enter the learning mode again in the future, you have to cycle the ignition on and off 4 times before starting the engine.

Start learning mode



Once the circuit board is in **learning mode**, the buzzer beeps and the LED flashes rhythmically every 1.5 seconds.
The engine can now be started and left idling
(Briefly press the throttle, otherwise there will be no RPM signal).

There is no time limit within which the engine must be started.

Once the engine is started, the buzzer stops only after the RPM signal is detected. The measurement takes place after an **8-second time delay**.
If there is no RPM signal, the buzzer will beep continuously!
Causes: Wiring errors, broken cables, etc...

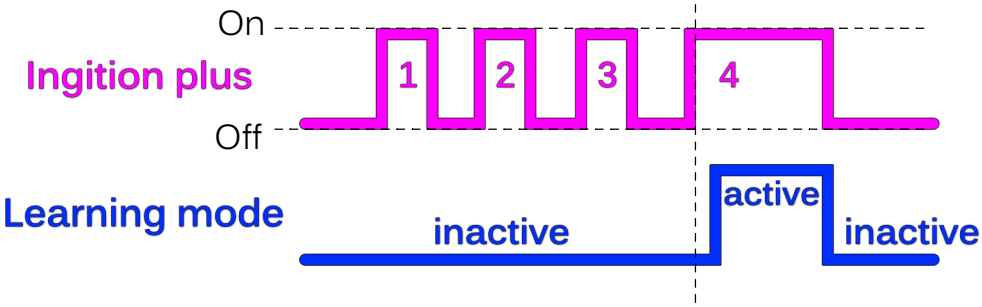
Once the engine has been started, the idle speed must be kept constant!

During the measurement, the average is calculated from several readings.

Another series of buzzer sounds, indicating that the current idle speed has been saved.

From now on, the saved values will be used to calculate the correct speed during normal operation.

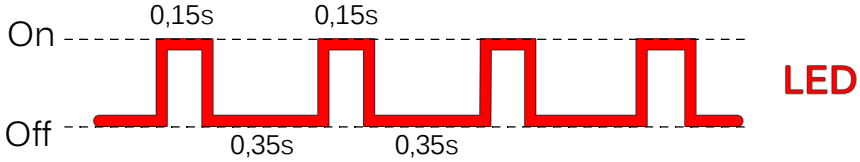
Abort learning mode



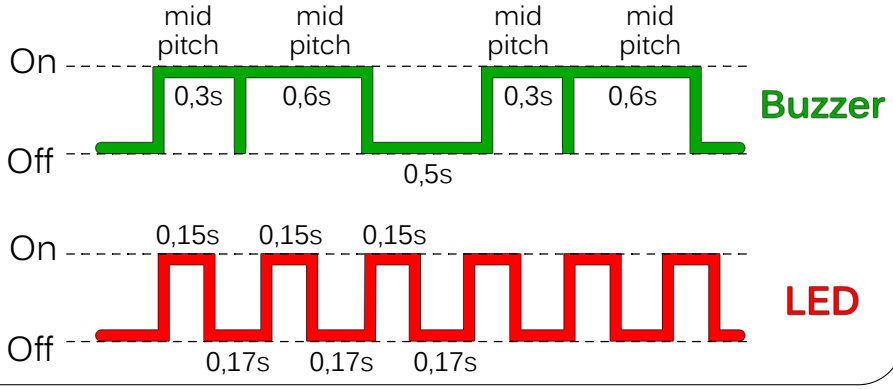
If you wish to abort the learning mode (the buzzer beeps rhythmically every 1.5 seconds), simply switch off the ignition without starting the engine.
Abortion is only possible if 'previous' values from a learning process have already been saved!

Function overview

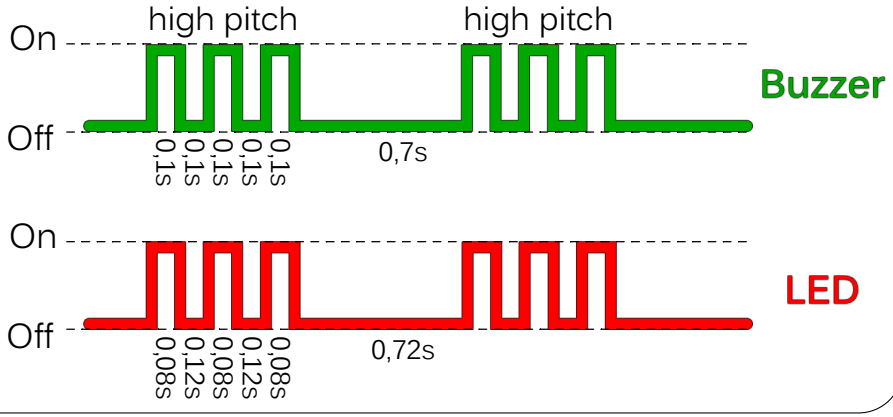
Oil pressure 1 too low



Oil pressure 2 too low and 2050 rpm exceeded



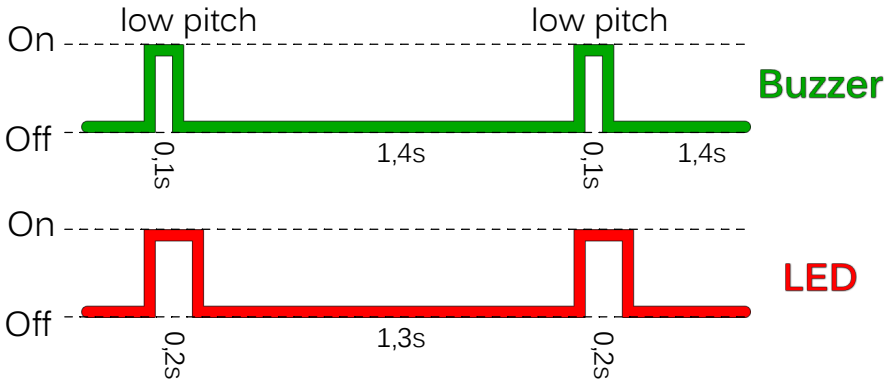
Oil pressure 2 OK / no RPM signal detected



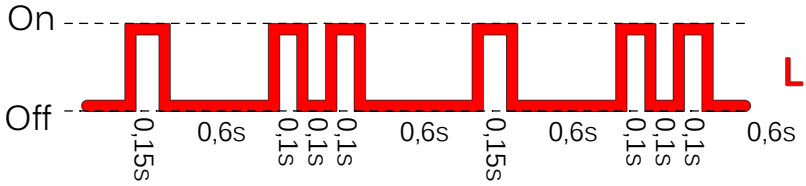
If a speed above 9000 rpm is detected, the buzzer chirps and the LED stays on!

Function overview in learning mode

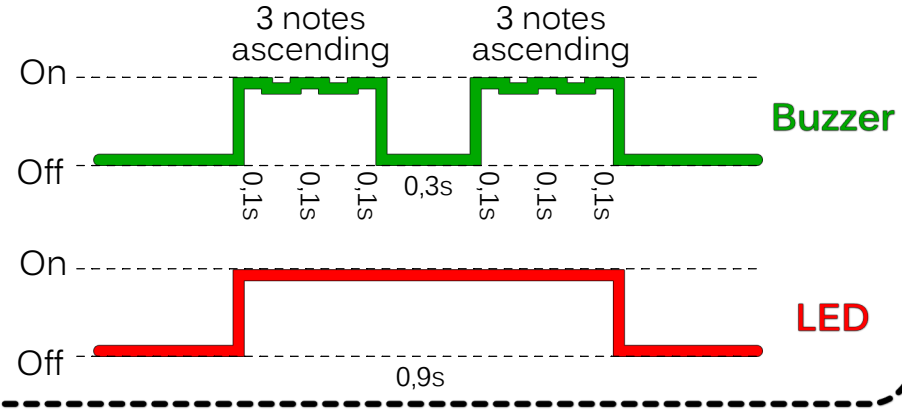
Learning mode (waiting for engine start)



Engine started (waiting for measurement)



Measurement successful, learning mode completed



This sequence is only once and doesn't repeat

Replaces the following dyn. oil pressure boards:

251 919 064 A	251 919 253 F
251 919 064 B	251 919 253 G
251 919 064 C	251 919 253 H
251 919 064 D	251 919 253 J
251 919 064 G	251 919 253 K
321 919 064 M	251 919 253 L

belt ratio (diesel):

2,75 2 belts without power steering (up to 1989)
2,39 Diesel with power steering (from 1986)
2,2 Diesel without power steering, single belt (from 1989)
2,01 Diesel with air conditioning, air conditioning and power steering, 2nd alternator or hydraulic pump

UART Interface

115200 / 8 data / 1 stop / no parity

The 3.3V UART output transmits information on the current RPM, all inputs and outputs, and the fault status every 50 ms.
When the software is started, all stored variables are displayed.

